ELECTRICAL ENGINEERING DEPARTMENT
COLORADO SCHOOL OF MINES

EENG 385 - Electronic Devices and Circuits
BJT Curve Tracer: Pseudo Ramp Generator
Lab Handout

Outcome and Objectives

The outcome of this lab is to analyze, simulate and assemble a circuit that generates a ramp-like waveform using an RC circuit and compare the expected behavior of the circuit from each mode of analysis. Through this process you will achieve the following learning objectives:

- Use a software tool to perform time and frequency domain analysis of an electronic circuit.
- Analyze and design a circuit containing resistors and op amps.
- Analyze and design a circuit consisting of several building blocks.
- Assemble a circuit on a PCB using the equipment in the laboratory.
- Use laboratory test and measurement equipment to analyze electronic circuits.

System Architecture

Today, we will explore the Pseudo Ramp Generator, a circuit generating the ramp-shaped voltage waveform shown in Figure 1. Note, the inclined portion of the voltage waveform, the ramp, is slightly curved. This slight curve, this deviation from a straight line, defines the "Pseudo" in Pseudo Ramp Generator.

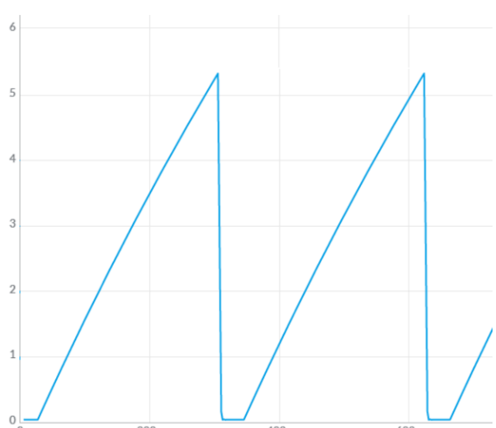

Figure 1: An example of the output the Pseudo Ramp Generator creates. Notice the slight curve in the ramp, this is result of charging a capacitor through a resistor.

We are now in our fourth lab analyzing, modeling, and constructing the various subsystems comprising the BJT Curve Tracer shown in Figure 2.

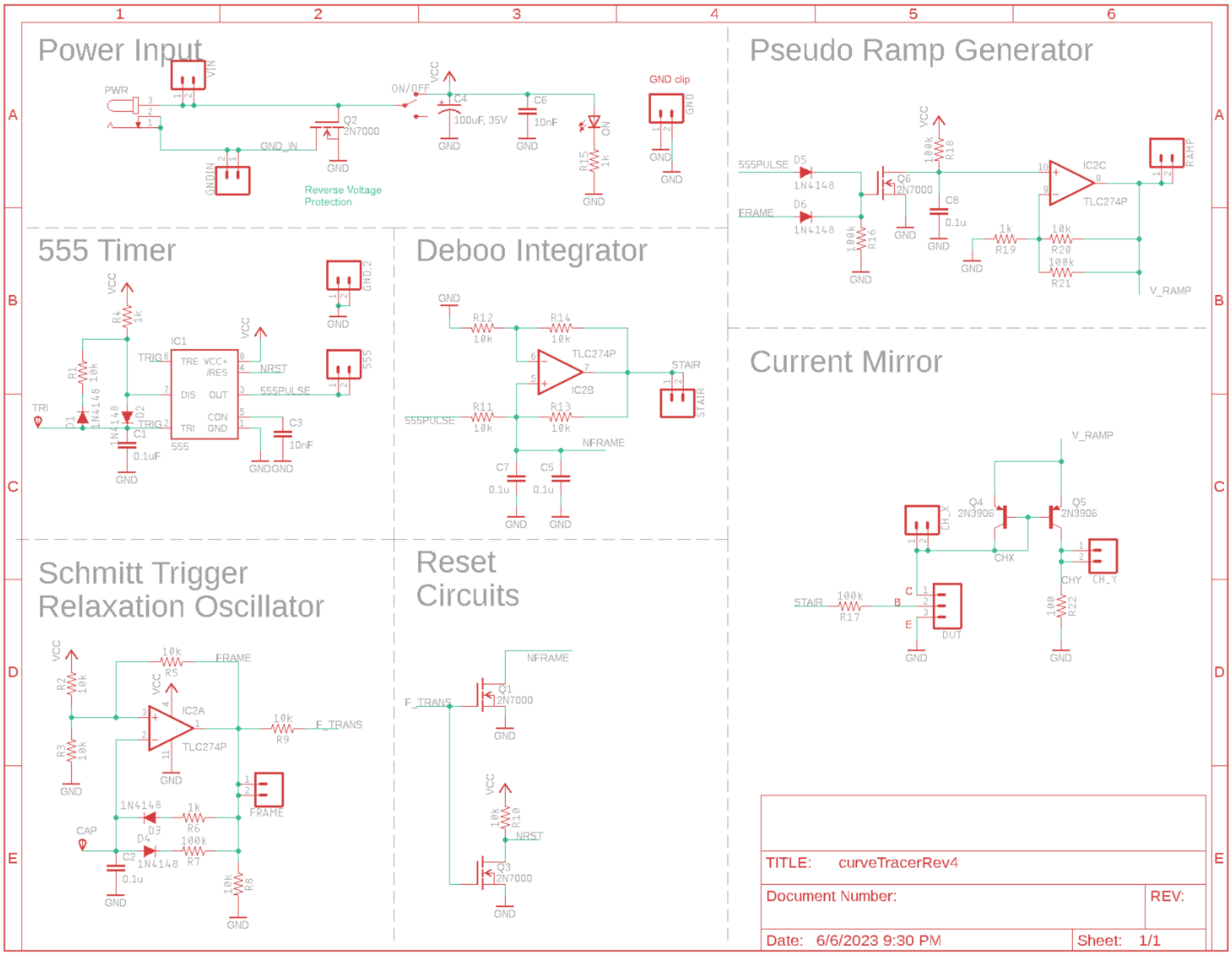

Figure 2: The complete BJT curve tracer.

Look carefully at the schematic shown in Figure 2 and locate the Pseudo Ramp Generator subsystem. Now, notice this subsystem is driven by the 555PULSE and FRAME signals. To understand the behavior of the Pseudo Ramp Generator, retrieve the simulation information you found in the previous labs and put it into Table 1.

Table 1: The output of the 555 Timer and Schmitt Trigger Oscillator simulations from the prior labs.

Quantity	555 Timer Simulation	Schmitt Trigger Relax Osc Simulation
Time high (μs)		
Time low (μs)		
Period (μs)		
Frequency (kHz)		
Duty Cycle		

We will come back to this table through the course of the lab.

Analysis: Pseudo Ramp Generator

When plotted as a function of time, a ramp function has a constant slope measured as $\frac{\Delta V}{\Delta t}$. The term "Pseudo" is introduced into the name because the circuit to be built, shown in Figure 3, does not have a constant slope; it changes slightly as shown in Figure 1. To understand this further, let's start the analysis of this circuit.

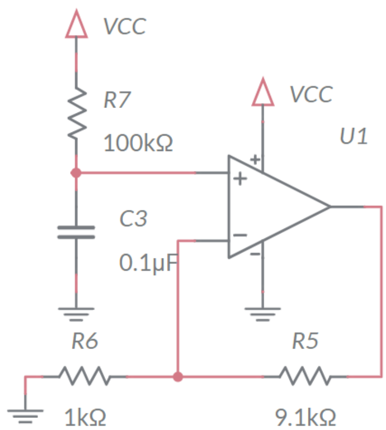

Figure 3: The Pseudo Ramp Generator circuit.

The analysis begins by recognizing the op amp configuration in Figure 3 is a non-inverting configuration. The resistors R5 and R6 form the feedback network. Note, the 9.1k resistor is formed from the parallel combination of 10k and 100k resistors. Why not just use a single 9.1k resistor? Two words, *inventory management* – it's easier to manage the inventory of 1k and 100k resistors that are used in all sorts of PCBs, then it is to stock a wide variety of odd-ball resistor values that used just once.

1. Determine the gain of the non-inverting op amp in Figure 3.
2. Determine the time constant for the RC network on the non-inverting input of the op amp in Figure 3. Remember that no current flows into an op amp input.
3. Write an equation describing the voltage of the charging capacitor. Assume V_{CC} = 9V.

So, the output of the op amp in Figure 3 is the familiar exponential charging of an RC circuit. But how on earth will this be used to generate the ramp shape shown in Figure 1? Well, if we limit the charging to the very beginning of the exponential function, then we get a signal that is very close to a straight line. To do this, we will need to limit the charging of the capacitor in Figure 3 to about 1V. We will do this by periodically removing the charge off the capacitor using a MOSFET just like we did with the Deboo Integrator.

Analysis: Pseudo Ramp Generator in BJT Curve Tracer

The Pseudo Ramp Generator will use a MOSFET to periodically remove the charge off the 0.1uF capacitor connected to the non-inverting input of the opamp. Figure 4 (left) shows the relevant portion of the Figure 2 schematic and Figure 4 (right) is a simplified

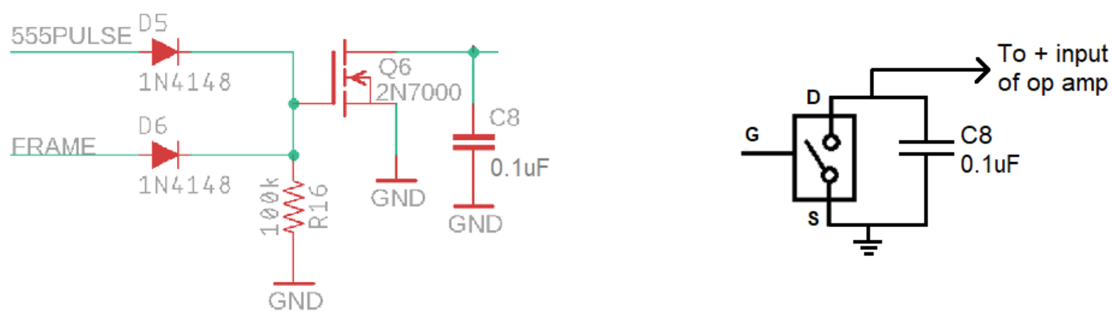

Figure 4: (left) The relevant portion of the circuit from Figure 2. (right) A simplified version of the circuit.

We will reset the charge on the Pseudo Ramp Generator's 0.1uF capacitor, using the MOSFET Q6 as a voltage-controlled switch, shown on the right-side of Figure 4. The behavior of this switch is the same as the previous lab, and given by:

- When the gate is driven towards 9V, the switch is closed.
  In other words, the drain **and** source are **connected**.

- When the gate is driven towards 0V, the switch is open.
  In other words, the drain **is disconnected** from the source.

Observe, the gate of the MOSFET in Figure 4 is driven by both the 555PULSE and FRAME signals through diodes. Let's take a moment to focus on the purpose of the two diodes D5, D6. In this analysis, you will use the ideal diode model which is defined by the following two bullet points.

- If the diode is forward biased (anode wants to be at a higher potential than the cathode), then the diode is a short circuit.
- If the diode is reverse biased (cathode wants to be at a higher potential than the anode), then the diode is an open circuit.

Use this model to determine the gate voltage and capacitor behavior for every combination of 555PULSE and FRAME voltage shown in Table 2. To help you get started, consider what happens in the FRAME=9V, 555PULSE=0V cell.

> *Diode D6 is forward biased because its anode is at 9V and the cathode is attached to ground through the resistor R16. Thus, the gate of the MOSFET is at 9V. This situation causes the MOSFET to act like a closed switch, connecting the drain of the MOSFET to the source. This closed switch causes any charge on the 0.1F capacitor to discharge. Diode D5 is reverse biased because its cathode is at a higher potential than its anode. So, diode D5 acts like an open circuit.*

Table 2: The behavior of the MOSFET when driven by the two diodes.

555PULSE\FRAME	FRAME = 0V	FRAME = 9V
555PULSE = 0V		Gate = 9V Capacitor is discharged
555PULSE = 9V		

The pair of diodes acts like a logic gate. Let 0V represent logic 0 and 9V represent logic 1.

1. What logic gate does the pair of diodes form when you consider 555PULSE and FRAME as the input and the MOSFET gate as the output?

Now let's put these ideas together into a picture of how the waveform generated by the Pseudo Ramp Generator will look in the simulator and when assembled.

2. During one period of the 555PULSE signal, how long will the 0.1uF capacitor in the Pseudo Ramp Generator by held at 0v? Assume the FRAME signal is at 0V.
3. During one period of the 555PULSE signal, how long will the 0.1uF capacitor in the Pseudo Ramp Generator be allowed to charge? Assume the FRAME signal is at 0V.
4. Assume that the 0.1uF capacitor is totally discharge. During one period of the 555PULSE signal, what is the highest voltage the 0.1uF capacitor will obtain? Assume the FRAME signal is at 0V.
5. Using the same assumptions as in the previous problem, what will be the highest voltage on the op amp output? *(Hint: Multiply the previous answer by the gain of the op amp found in an earlier question.)* Put this answer in the Analysis column of Table 3 at the end of the lab.

The reason the FRAME signal in this circuit is because the FRAME pulse resets the 555 Timer via the NRST signal. This 555 reset stops the 555 Timer from generating pulses and consequently stops the 555PULSE signal from clearing the charge off the 0.1uF capacitor. So, when the 555 Timer is in reset due to the FRAME signal, the FRAME signal is responsible for clearing the charge off the 0.1uF capacitor.

So now we know how much of the charging exponential function we are using to approximate a straight line. The next question to address becomes: Is this part of the exponential charging of the capacitor straight enough? Let's examine the question further by simulating it.

Simulation: Pseudo-Ramp Generator

Use LTspice to build and simulate the Schmitt Trigger relations oscillator circuit shown in Figure 5. Enter the highest voltage produced on the RAMP in the **Simulation** column of Table 3 at the end of the lab. Make sure to save a copy of your schematic and timing windows for your lab report.

Net Names	trig, 555out, Vcc, ramp
Component	NE555, T2N7002 LT1014
Waveform Nets	ramp, 555out
Configure Analysis	Transient, Stop Time 20ms

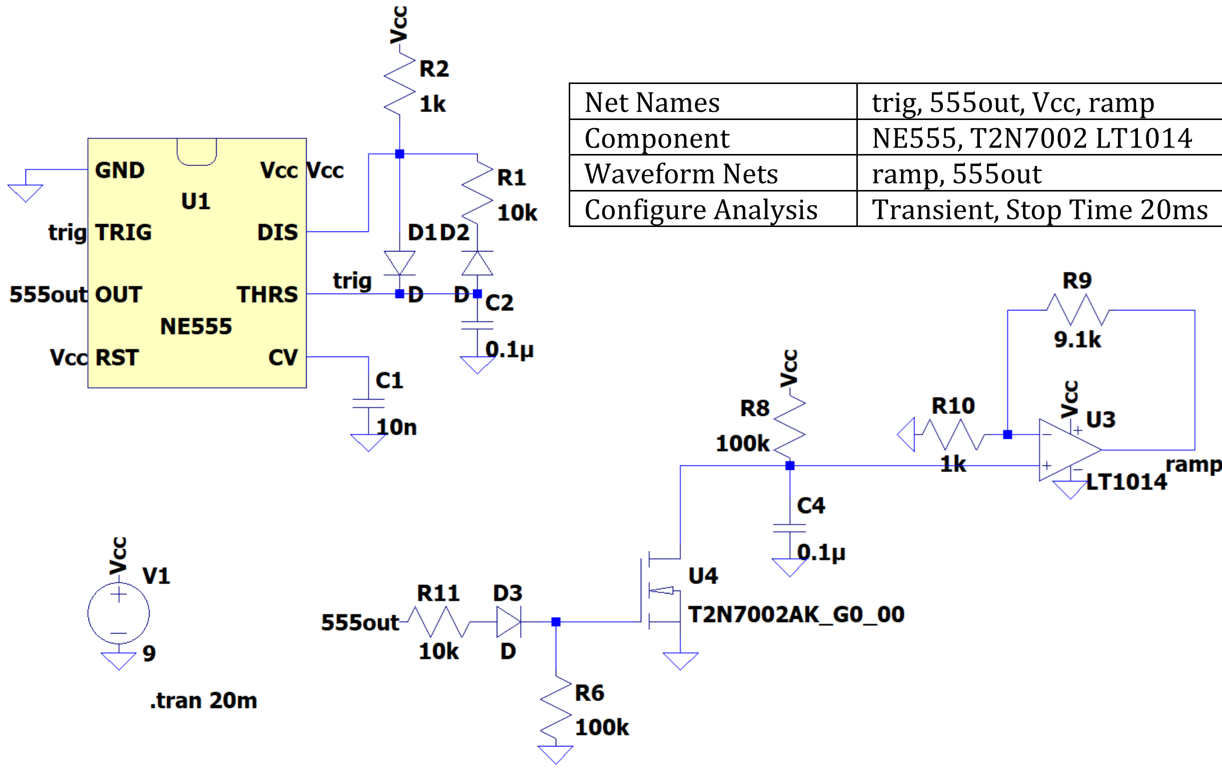

Figure 5: The circuit used in the simulation does not include the Schmitt Trigger Relaxation Oscillator.

Empirical: Pseudo Ramp Generator

Once you have completed assembly of your PSEUDO RAMP GENERATOR subsystem, perform the following test.

1) Power up an oscilloscope, attach a probe to Channel 1 and configure it as follows.

Ch1 probe	555 test point
Ch1 ground clip	GND test point
Horizontal (scale)	500us
Ch1 (scale)	2V
Ch2 probe	RAMP test point
Ch2 (scale)	Same as Channel 1
Trigger mode	Auto
Trigger source	Ch1
Trigger slope	↑
Trigger level	4.5V

2) Set the GND reference of Ch1 and Ch2 to the lowest visible reticule. The waveforms will overlap the same as they did in the simulation. Set the horizontal position of the trigger to the left most visible reticule. Take a screen shot of this output and include it in your lab report.

3) Take a screen shot of the of the 555PULSE and RAMP waveforms and include them in your lab report. Screen shot the oscilloscope traces on USB. Cell phone pictures will lose points. [Save] → Save → Format → 24-bit Bit... (*.bmp) [Save] → Save → Press to Save

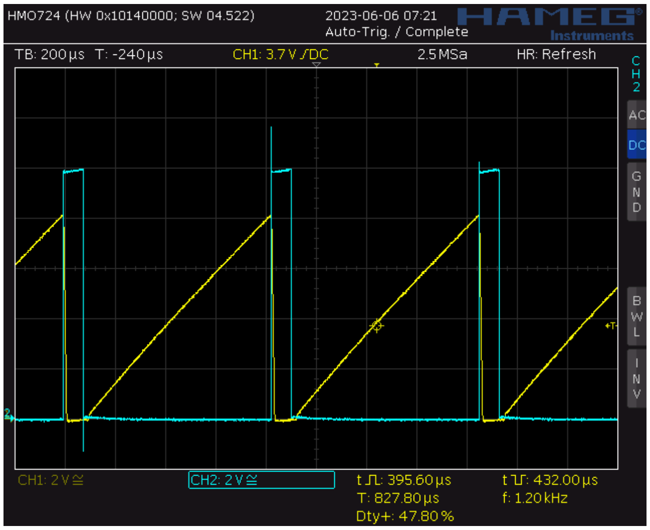

Figure 6: An oscilloscope trace showing the two output you need to capture. Note that this image was captured on a Rhode&Schwarz HMO724.

Comparison: Pseudo Ramp Generator

Complete the **Analysis, Simulation** and **Empirical** columns of the following tables using the information you found throughout this lab. Represent your answer to 3 significant figures using the units given in parenthesis in the **Quantity** column. You will need this table in later labs, so keep it handy.

Table 3: Summary of the voltage produced by the Pseudo Ramp Generator.

Quantity	Analysis	Simulation	Empirical
Ramp Amplitude (volts)			

Turn In: Pseudo Ramp Generator

1) Make a record of your response to numbered items below and turn in a single copy as your team's solution on Canvas using the instructions posted there.
2) Include the names of both team members at the top of your solutions.
3) Use complete English sentences to introduce what each of the items listed below is and how it was derived.

Hint, use Ctrl+click to follow links. This also works for all the Figures and Tables in these labs.

Analysis: Pseudo Ramp Generator

Steps 1 – 3 of analysis

Analysis: Pseudo Ramp Generator in BJT Curve Tracer

Table 2

Steps 1 – 5 of analysis

Simulation: Pseudo Ramp Generator

Schematic

Waveform

Assemble: Pseudo Ramp Generator

Screen shot oscilloscope output for 555PULSE and RAMP.

Comparison: Pseudo Ramp Generator

Table 3 Compare Pseudo Ramp Generator output in different models.